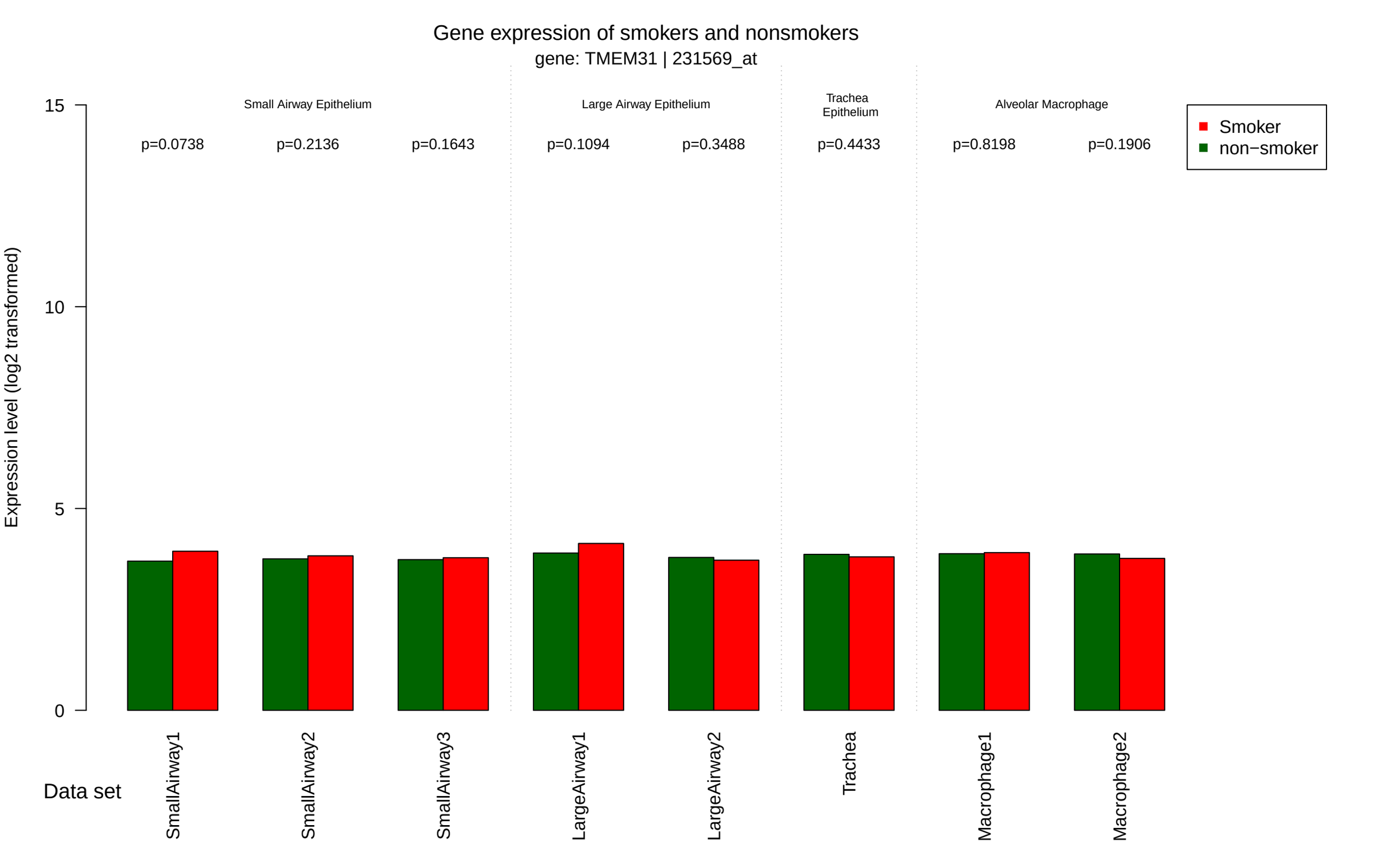
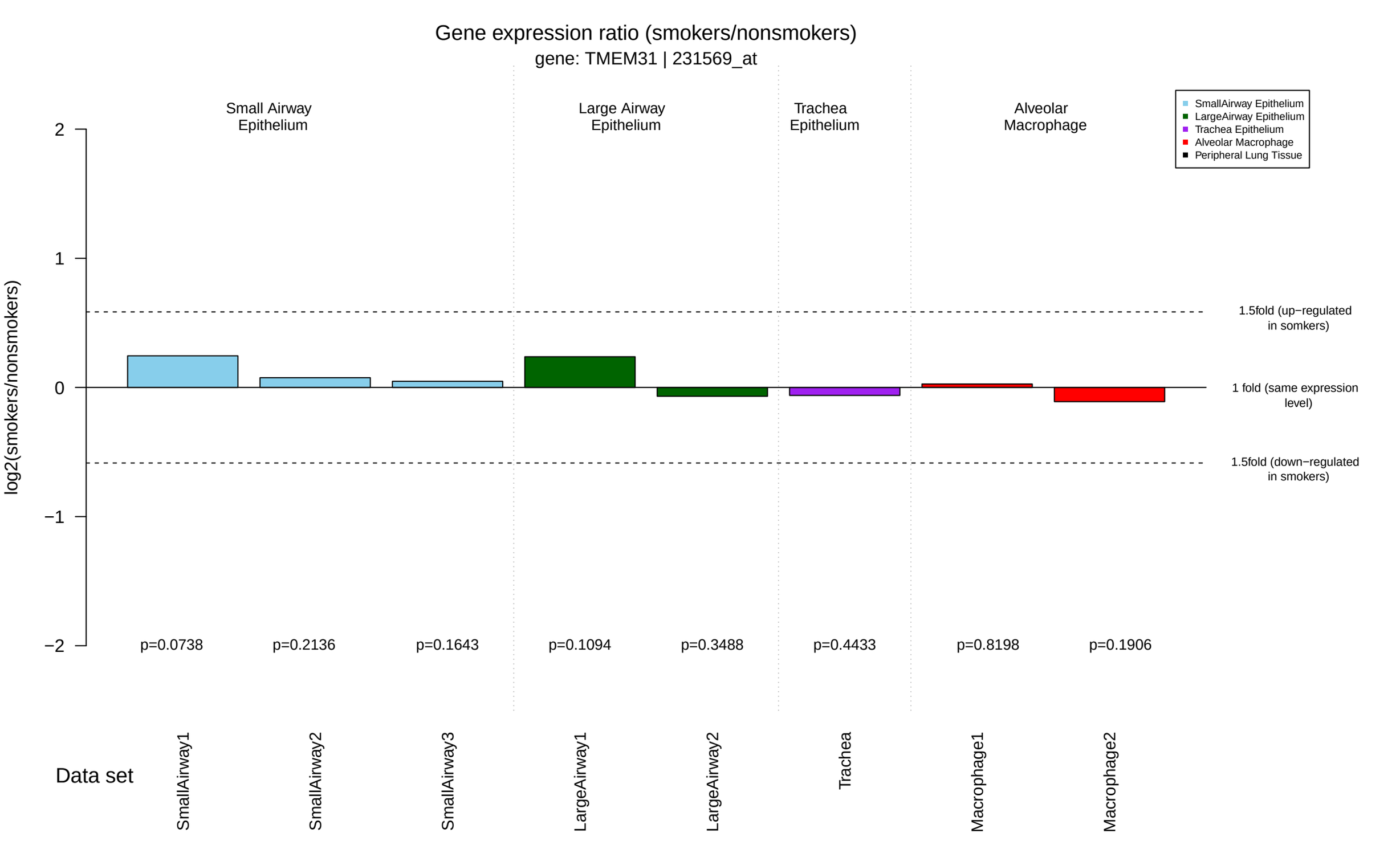
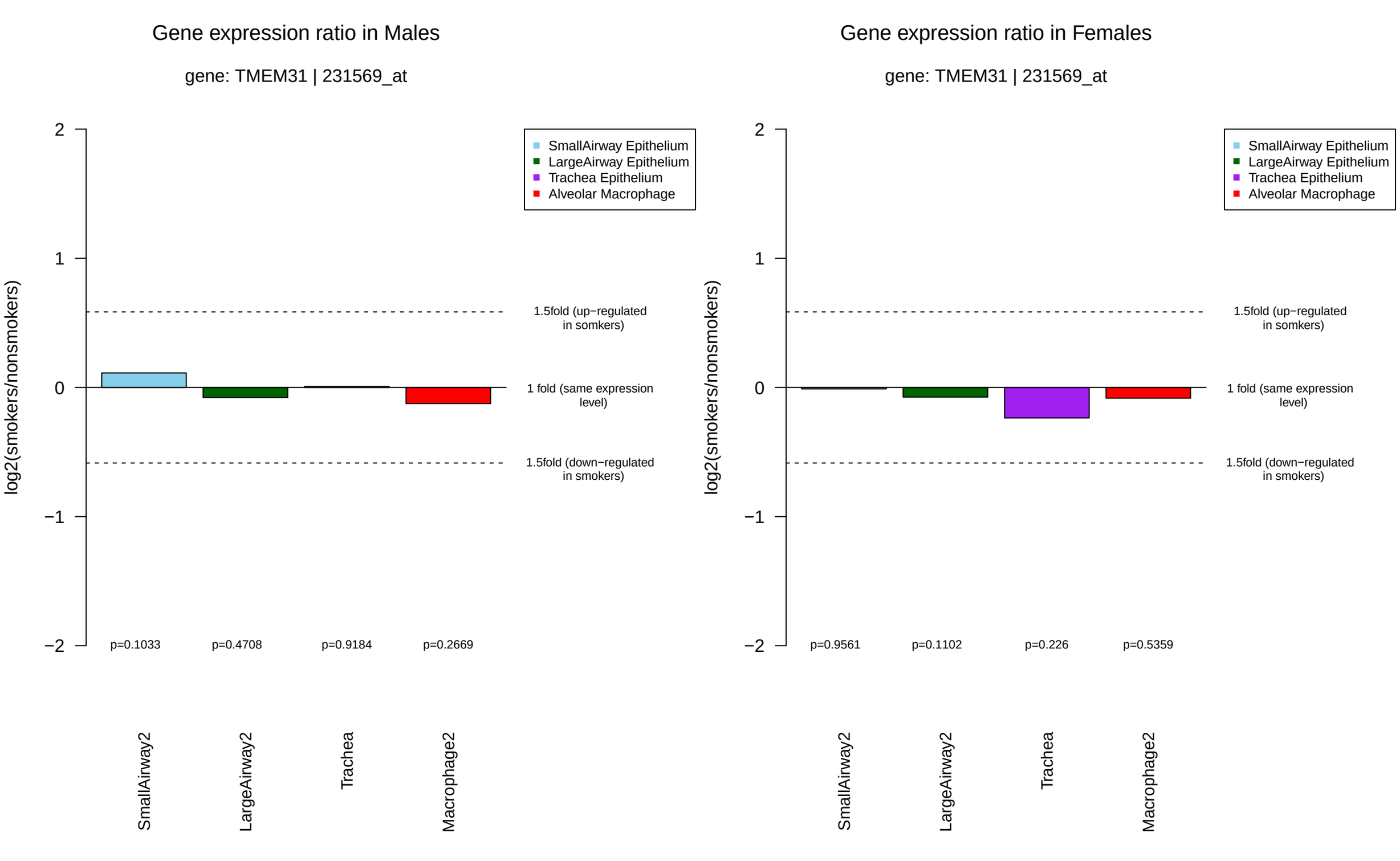
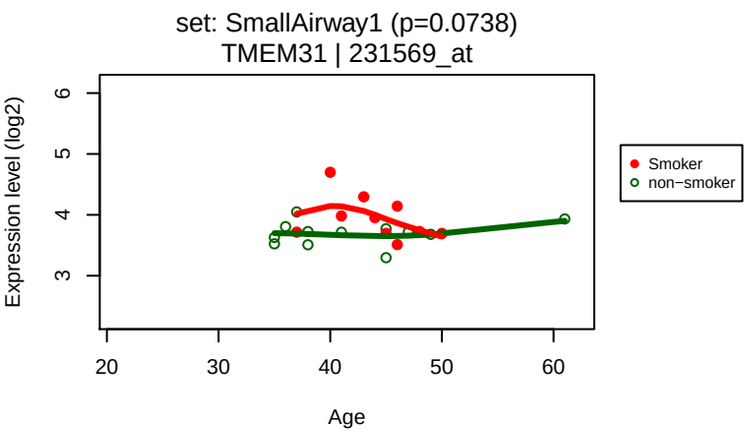
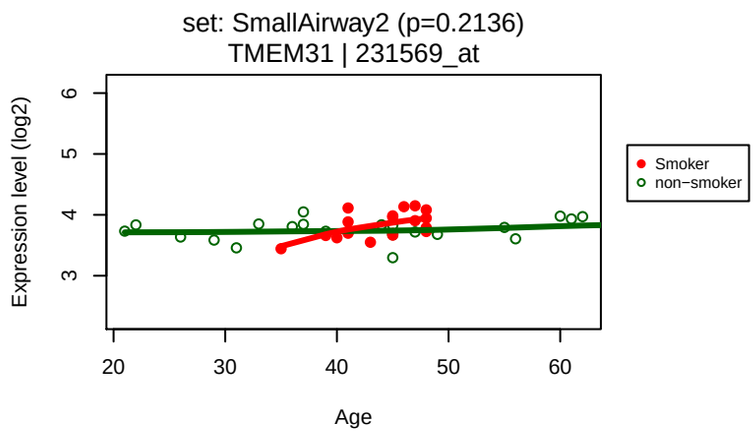
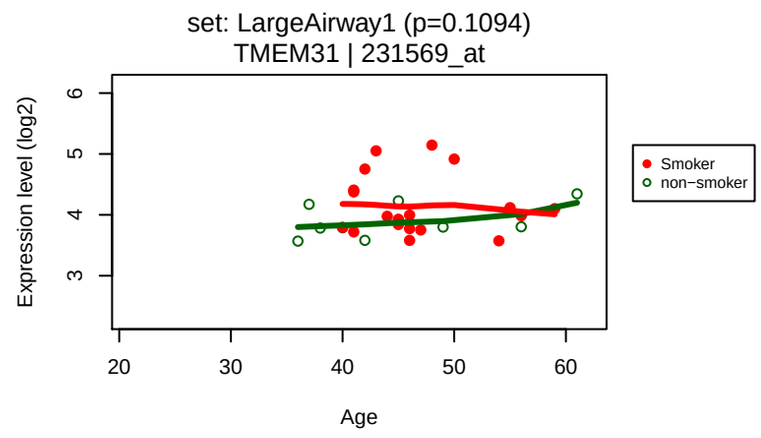

Gene expression of smokers and nonsmokers
gene: TMEM31 | 231569_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0738
p=0.2136
p=0.1643
p=0.1094
p=0.3488
p=0.4433
p=0.8198
p=0.1906
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

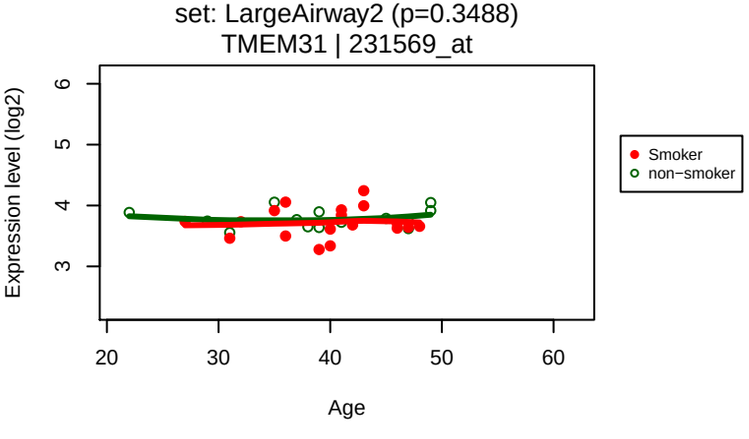

Gene expression ratio (smokers/nonsmokers)
gene: TMEM31 | 231569_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0738
p=0.2136
p=0.1643
p=0.1094
p=0.3488
p=0.4433
p=0.8198
p=0.1906
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

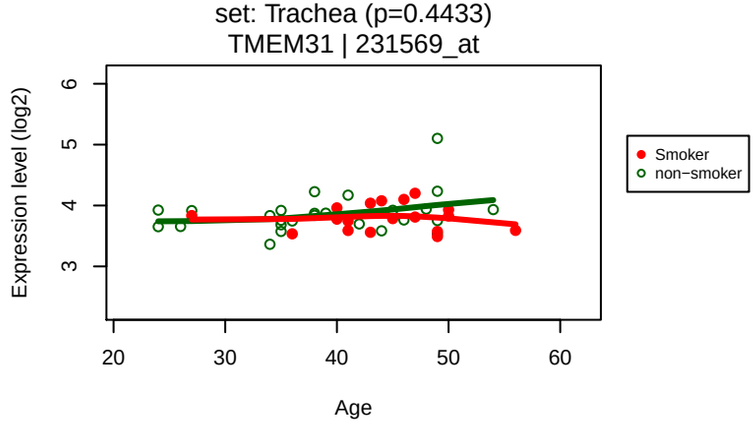

Gene expression ratio in Males
gene: TMEM31 | 231569_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1033
p=0.4708
p=0.9184
p=0.2669
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM31 | 231569_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9561
p=0.1102
p=0.226
p=0.5359
SmallAirway2
LargeAirway2
Trachea
Macrophage2

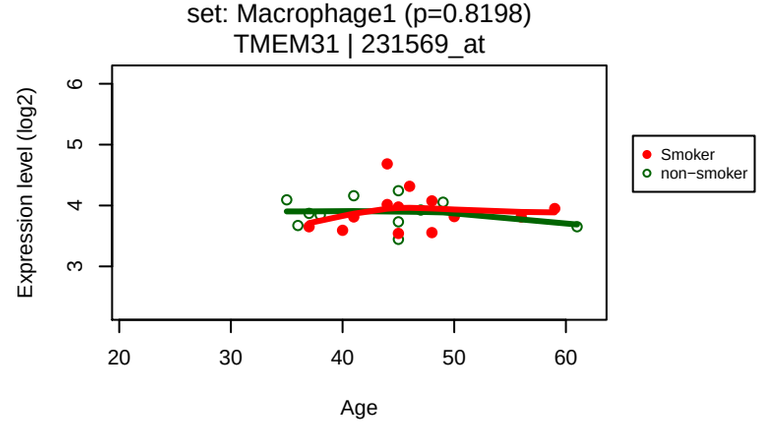

set: SmallAirway1 (p=0.0738)
TMEM31 | 231569_at
Expression level (log2)
Age
Smoker
non−smoker

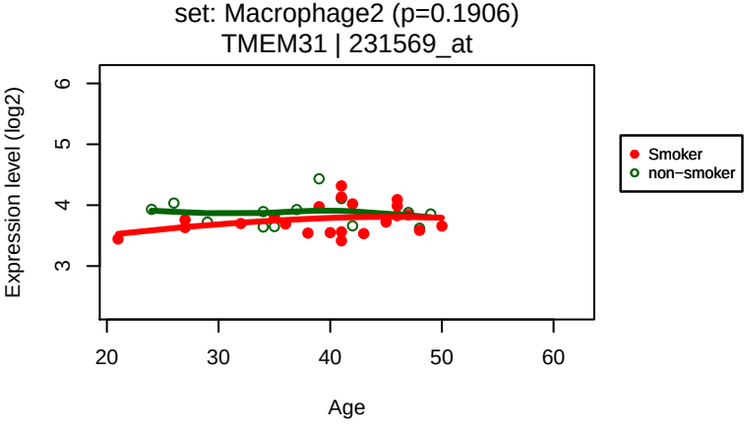

set: SmallAirway2 (p=0.2136)
TMEM31 | 231569_at
Expression level (log2)
Age
Smoker
non−smoker

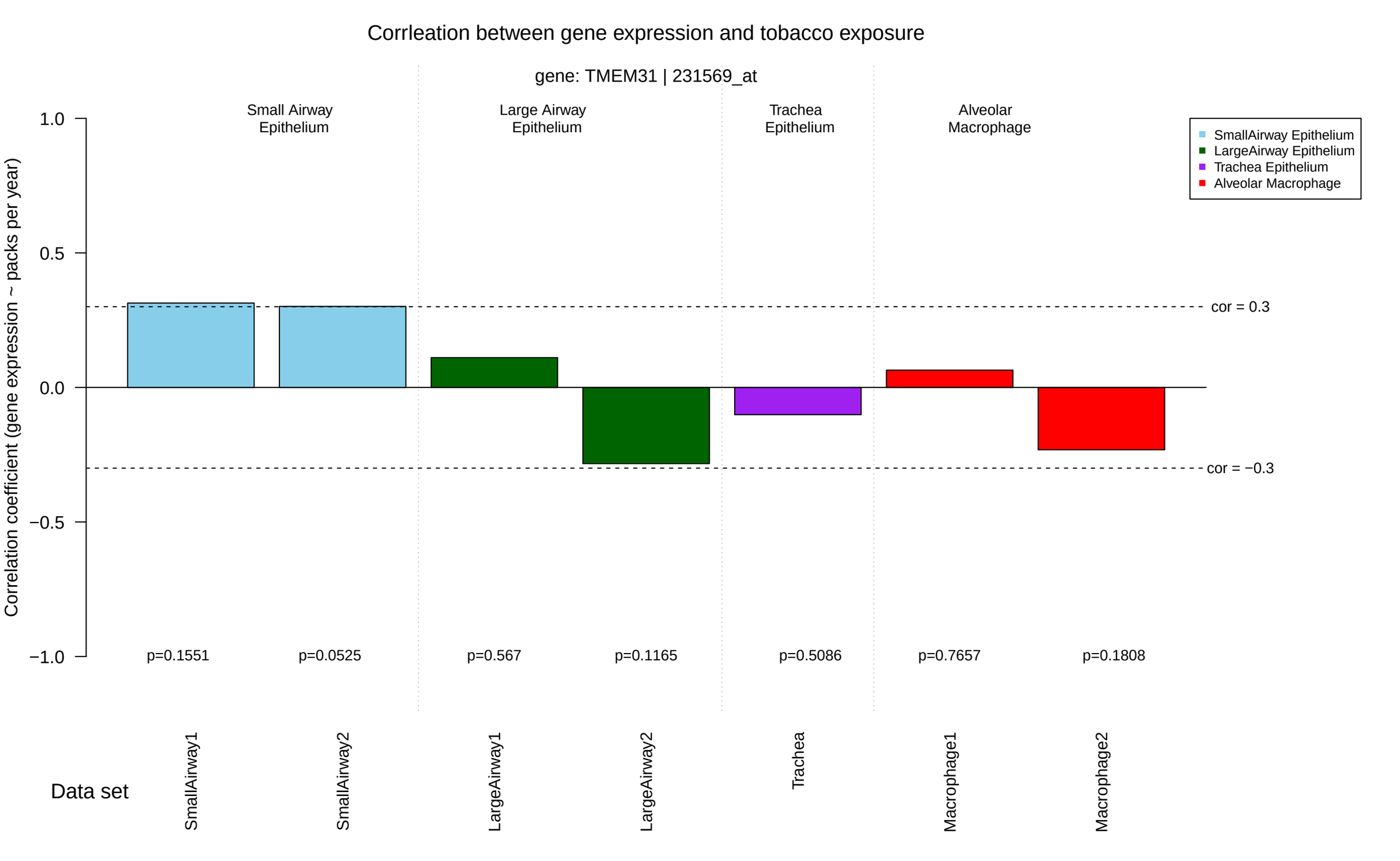

set: LargeAirway1 (p=0.1094)
TMEM31 | 231569_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3488)
TMEM31 | 231569_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4433)
TMEM31 | 231569_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8198)
TMEM31 | 231569_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1906)
TMEM31 | 231569_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM31 | 231569_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1551
p=0.0525
p=0.567
p=0.1165
p=0.5086
p=0.7657
p=0.1808
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2